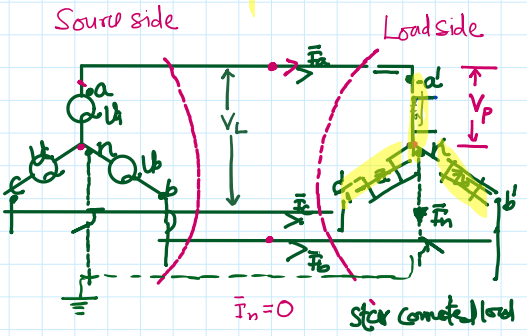
Lecture 24: Three-phase Systems (Contd...)

Line and Phase Quantities

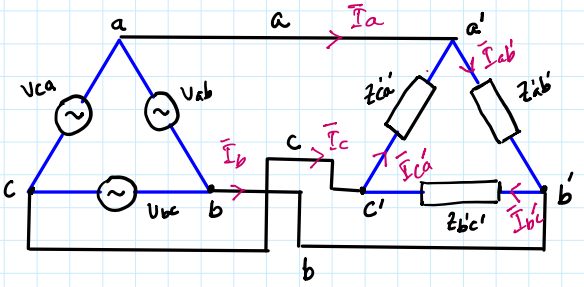
The load in three-phase system may be connected in star or delta as shown below. Accordingly we have line and phase quantities as shown in figure below.

Y - Y
Y - Δ
Δ - Y
Δ - Δ

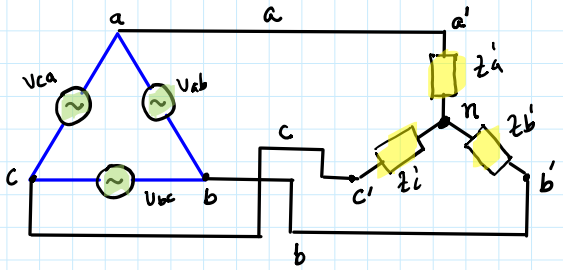


Star-star Configuration.

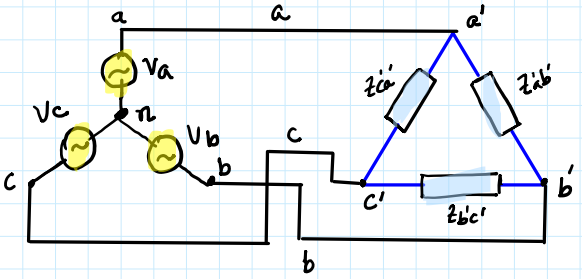
Each configuration has phase and line, voltage and current quantities. And there is relationship between the phase and line quantities of each configuration.



Delta-delta Configuration



Delta-Star Configuration



Star-delta Configuration

Line and Phase Quantity Relationship:

(A) Star Connected load: Phase quantities:

$$\bar{V}_{an} = V \angle 0$$
$$\bar{V}_{bn} = V \angle -120^\circ$$
$$\bar{V}_{cn} = V \angle 120^\circ$$

$$Z_a = Z_b = Z_c = Z$$

Line Currents and Phase Currents:

$$\bar{I}_a = \frac{\bar{V}_a}{Z_a} = \frac{V}{|Z|}\underline{|-\phi_z} = I\underline{|-\phi}$$

$$\bar{I}_b = \frac{\bar{V}_b}{Z_b} = \frac{V\underline{|-120^\circ}}{|Z|\underline{|\phi_z}} = \frac{V}{Z}\underline{|-120^\circ-\phi} = I\underline{|-120^\circ-\phi}$$

$$\bar{I}_c = \frac{\bar{V}_c}{Z_c} = \frac{V\underline{|120^\circ}}{|Z|\underline{|\phi_z}} = \frac{V}{|Z|}\underline{|120^\circ-\phi} = I\underline{|120^\circ-\phi}$$

These currents are called phase currents as well as line currents.

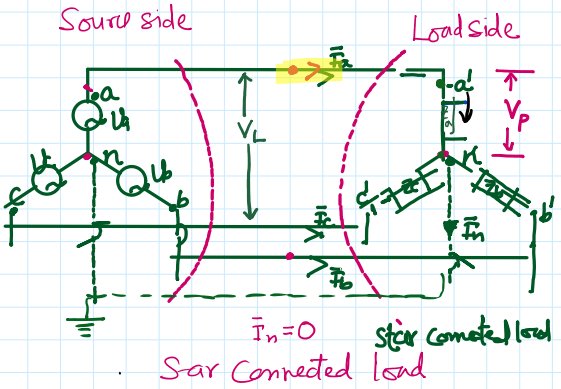
So for Y connected load line currents = Phase currents.

$$\boxed{I_L = I_p = I}$$

Line Voltages and Phase Voltages:

Line voltage = Line to Line voltages

= $\bar{V}_{ab}, \bar{V}_{bc}, \bar{V}_{ca}$ are line to line voltages



Star Connected load

$$\bar{V}_{ab} = \bar{V}_a - \bar{V}_b$$

$$= V\underline{|0} - V\underline{|-120^\circ}$$

$$= \sqrt{3}V\underline{|30^\circ}$$

$$v_{ab}(t) = v_a(t) - v_b(t)$$

$$= \sqrt{2}V\sin\omega t - \sqrt{2}V\sin(\omega t - 120^\circ)$$

$$= \sqrt{3}\sqrt{2}V\sin(\omega t + 30^\circ)$$

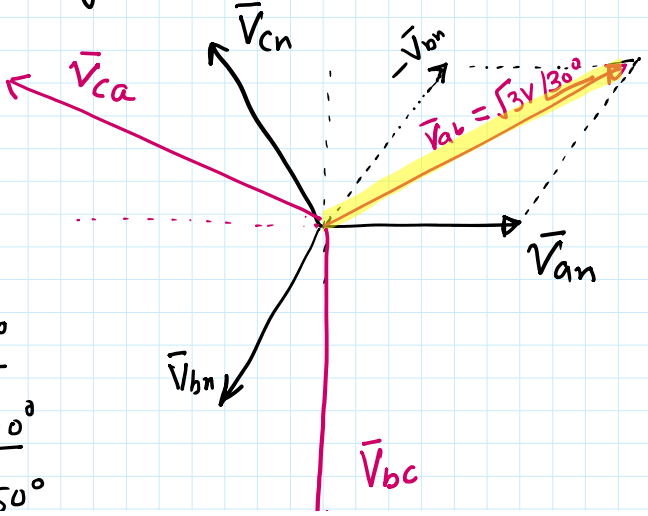
$$v_{ab}(t) = \sqrt{2}V_{LL}\sin(\omega t + 30^\circ)$$

where V_{LL} = RMS value of line to line voltages

Similarly we can write,

$$\bar{V}_{bc} = \sqrt{3}V\underline{|-90^\circ}$$

$$\bar{V}_{ca} = \sqrt{3}V\underline{|150^\circ}$$



Thus, we have,

$$\bar{V}_{ab} = \sqrt{3}V\underline{|30^\circ} = V_{LL}\underline{|30^\circ}$$

$$\bar{V}_{bc} = \sqrt{3}V\underline{|-90^\circ} = V_{LL}\underline{|-90^\circ}$$

$$\bar{V}_{ca} = \sqrt{3}V\underline{|150^\circ} = V_{LL}\underline{|150^\circ}$$

$$\bar{V}_{bc} = \sqrt{3}V \angle -90^\circ = V_{LL} \angle -90^\circ$$

$$\bar{V}_{ca} = \sqrt{3}V \angle 150^\circ = V_{LL} \angle 150^\circ$$

$\bar{V}_{bc}$ ↓

where V_{LL} = RMS value of line to line voltage $= \sqrt{3}\,V$

$V_a = V_b = V_c = V = V_p$ called phase voltage

Thus, line to line and phase voltage are related as

$$\boxed{V_L = V_{L-L} = \sqrt{3}\,V_p}$$

Powers: $\bar{S} = \bar{V}_a \bar{I}_a^* + \bar{V}_b \bar{I}_b^* + \bar{V}_c \bar{I}_c^*$

$= 3\bar{V}_a \bar{I}_a^* = 3V(I\angle -\phi)^* = 3VI\angle\phi$

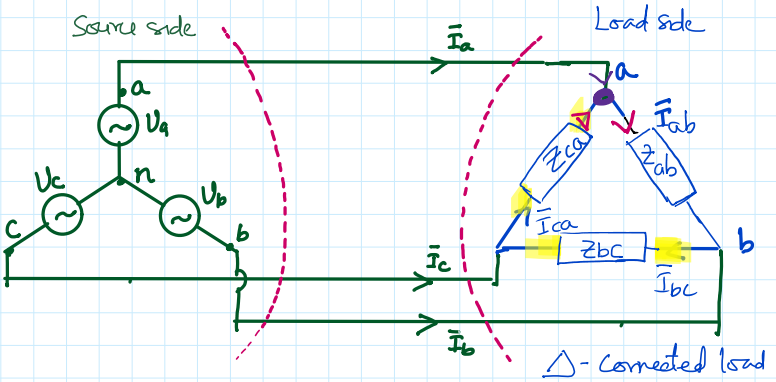
$= 3VI\cos\phi + j\,3VI\sin\phi$

$\bar{S} = P + jQ \Rightarrow P = 3VI\cos\phi$

$Q = 3VI\sin\phi$

$pf = P/S = \cos\phi$

Delta-Connected Load:



Loads are connected between ab, bc and ca terminals.

Hence $\bar{I}_{ab}$, $\bar{I}_{bc}$, $\bar{I}_{ca}$ are called as load currents or phase currents.

and phase voltages are $\bar{V}_{ab}$, $\bar{V}_{bc}$, $\bar{V}_{ca}$ which are line to line voltages.

Let $Z_{ab} = Z_{bc} = Z_{ca} = Z$ (balanced load)

$\bar{V}_{an}$, $\bar{V}_{bn}$, $\bar{V}_{cn}$ are source voltages for phase-a, b, c respectively.

Then $\bar{V}_{ab} = \sqrt{3}V\angle 30^\circ$, $\bar{V}_{bc} = \sqrt{3}V\angle -90^\circ$, $\bar{V}_{ca} = \sqrt{3}V\angle 150^\circ$ V

are line to line as well as phase voltages.

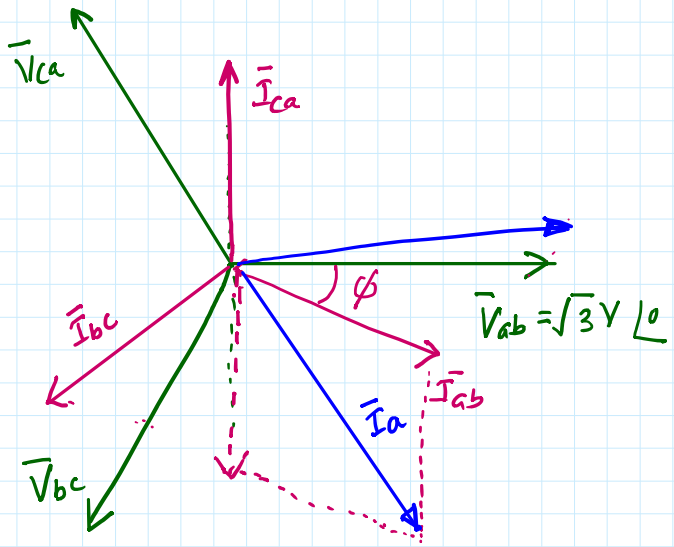
Therefore; $\boxed{V_{LL} = V_p = \sqrt{3}V}$

Line and Phase Currents: $\bar{I}_a, \bar{I}_b, \bar{I}_c$ are line currents

$\bar{I}_{ab}, \bar{I}_{bc}, \bar{I}_{ca}$ are load or phase currents

from the circuit: $\bar{I}_a = \bar{I}_{ab} - \bar{I}_{ca}$

$\bar{I}_a = I\angle -\phi - I\angle 120^\circ - \phi = \sqrt{3}I\angle -30^\circ - \phi$



Assuming $\bar{V}_{ab}$ as reference voltage

$\bar{V}_{ab} = \sqrt{3}V\angle 0$

$\bar{V}_{bc} = \sqrt{3}V\angle -120^\circ$

$\bar{V}_{ca} = \sqrt{3}V\angle 120^\circ$

$\bar{I}_{ab} = I\angle -\phi$

$\bar{I}_{bc} = I\angle -120^\circ - \phi$

$\bar{I}_{ca} = I\angle 120^\circ - \phi$

Similarly $\bar{I}_b = \bar{I}_{bc} - \bar{I}_{ab} = I\angle -120^\circ - \phi - I\angle -\phi$

$= \sqrt{3}I\angle -150^\circ - \phi$

$\bar{I}_c = \bar{I}_{ca} - \bar{I}_{bc} = I\angle 120^\circ - \phi - I\angle -120^\circ - \phi$

$= \sqrt{3}I\angle 90^\circ - \phi$

Thus we see line currents I_a, I_b, I_c are $\sqrt{3}\, I_{phase}$

$$I_a = I_b = I_c = I_L = \sqrt{3}\, I_p$$